

SVM Notes: Derivations, Kernels, and Examples

Geometric Derivation of the Margin Width

Goal: Prove that the width of the margin is $\frac{2}{\|W\|}$.

Defining the Hyperplanes

We define the decision boundary and the two "gutters" (margins) as follows:

$$H_0 : W \cdot X + B = 0 \quad (\text{Decision Surface}) \quad (1)$$

$$H_1 : W \cdot X + B = 1 \quad (\text{Positive Support Vector Plane}) \quad (2)$$

$$H_2 : W \cdot X + B = -1 \quad (\text{Negative Support Vector Plane}) \quad (3)$$

Calculating Distance

Let X_+ be a point on H_1 and X_- be a point on H_2 . To find the distance (width) between H_1 and H_2 , we project the vector $(X_+ - X_-)$ onto the unit normal vector of the hyperplane, which is $\frac{W}{\|W\|}$.

$$\text{Width} = (X_+ - X_-) \cdot \frac{W}{\|W\|}$$

Substituting the plane equations ($W \cdot X_+ = 1 - B$ and $W \cdot X_- = -1 - B$):

$$\begin{aligned} \text{Width} &= \frac{W \cdot X_+ - W \cdot X_-}{\|W\|} \\ &= \frac{(1 - B) - (-1 - B)}{\|W\|} \\ &= \frac{1 - B + 1 + B}{\|W\|} \\ &= \frac{2}{\|W\|} \end{aligned}$$

Conclusion

To maximize the margin width $\frac{2}{||W||}$, we must **minimize** the norm of the weights $||W||$.

2. The Optimization Problem (Primal Form)

Minimizing $||W||$ is mathematically equivalent to minimizing $\frac{1}{2}||W||^2$. We must also ensure all data points are correctly classified outside the margin.

Objective:

$$\min_{W, B} \frac{1}{2} ||W||^2$$

Subject to Constraints:

$$y_i(W \cdot x_i + B) \geq 1 \quad \forall i$$

3. The Lagrangian Derivation (Dual Form)

To solve this constrained optimization, we use Lagrange Multipliers (α_i). We construct the Lagrangian function L :

$$L(W, B, \alpha) = \frac{1}{2} ||W||^2 - \sum_{i=1}^n \alpha_i [y_i(W \cdot x_i + B) - 1]$$

To find the optimum, we take partial derivatives with respect to W and B and set them to 0.

Derivatives

1. With respect to W :

$$\frac{\partial L}{\partial W} = W - \sum_{i=1}^n \alpha_i y_i x_i = 0 \implies W = \sum_{i=1}^n \alpha_i y_i x_i$$

2. With respect to B :

$$\frac{\partial L}{\partial B} = - \sum_{i=1}^n \alpha_i y_i = 0 \implies \sum_{i=1}^n \alpha_i y_i = 0$$

Final Dual Formulation

Substituting the expression for W back into the Lagrangian yields the Dual form, which depends only on the dot product of data points:

$$\max_{\alpha} \left[\sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (x_i \cdot x_j) \right]$$

The Kernel Trick

Standard SVMs are linear classifiers. The Kernel Trick allows them to solve non-linear problems (like concentric circles) efficiently.

The Problem: Non-Linear Data

In 2D, some datasets cannot be separated by a straight line. For example, if red dots are in the center and blue dots surround them, no single line works.

The Solution: Higher Dimensions

We can project the data into a higher-dimensional space (mapping function $\phi(x)$) where it becomes linearly separable.

- **Analogy:** If red and blue balls are flat on a table (2D), we can lift the red balls up (3D). Now, a flat sheet can separate the floating red balls from the blue ones below.

The "Trick": Computational Efficiency

Calculating the explicit mapping $\phi(x)$ for high dimensions is computationally expensive. However, notice that the Dual Formulation (Section 3) **only** requires the dot product $(x_i \cdot x_j)$.

We define a **Kernel Function** K such that:

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

This calculates the high-dimensional dot product directly from the original data, skipping the explicit mapping step.

Mathematical Proof (Example)

Consider mapping 2D vectors $x = (x_1, x_2)$ and $z = (z_1, z_2)$ to 3D using a squaring mapping.

Method A: The Hard Way (Explicit Mapping) Mapping: $\phi(x) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$.
Dot Product $\phi(x) \cdot \phi(z)$:

$$= x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2$$

Method B: The Kernel Trick Using polynomial kernel $K(x, z) = (x \cdot z)^2$:

$$(x_1 z_1 + x_2 z_2)^2 = x_1^2 z_1^2 + 2x_1 x_2 z_1 z_2 + x_2^2 z_2^2$$

Conclusion: Method B gives the exact same result as Method A but with much simpler math.

Common Kernels

- **Polynomial:** $K(x, z) = (x \cdot z + c)^d$. Good for curved boundaries.
- **RBF (Gaussian):** $K(x, z) = \exp(-\gamma \|x - z\|^2)$. Maps to infinite dimensions; highly flexible.

Worked Example: Full Solution

Problem Statement

Given:

$$x_1 = 1, y_1 = +1 \quad x_2 = 3, y_2 = -1$$

Kernel: Polynomial, degree $d = 1$: $k(x_i, x_j) = x_i x_j + 1$

Regularisation: $C = 5$

Find: (i) Optimal α_1, α_2 ; (ii) discriminant function $f(z)$; (iii) output class for $z = 2$.

Step 1 — Set Up the Dual Objective

The dual maximisation with $n = 2$ is:

$$\max_{\alpha_1, \alpha_2} \left[\sum_{i=1}^2 \alpha_i - \frac{1}{2} \sum_{i=1}^2 \sum_{j=1}^2 \alpha_i \alpha_j y_i y_j (x_i x_j + 1) \right]$$

Expanding all four (i, j) pairs:

$$= \max \left[(\alpha_1 + \alpha_2) - \frac{1}{2} \left(\alpha_1^2 y_1^2 (x_1 x_1 + 1) + \alpha_1 \alpha_2 y_1 y_2 (x_1 x_2 + 1) \right. \right. \\ \left. \left. + \alpha_2 \alpha_1 y_2 y_1 (x_2 x_1 + 1) + \alpha_2^2 y_2^2 (x_2 x_2 + 1) \right) \right]$$

Substituting $x_1 = 1$, $x_2 = 3$, $y_1 = 1$, $y_2 = -1$:

$$\begin{aligned} k(x_1, x_1) &= 1 \cdot 1 + 1 = 2, & k(x_1, x_2) &= 1 \cdot 3 + 1 = 4, \\ k(x_2, x_1) &= 3 \cdot 1 + 1 = 4, & k(x_2, x_2) &= 3 \cdot 3 + 1 = 10 \end{aligned}$$

$$\begin{aligned} &= \max \left[(\alpha_1 + \alpha_2) - \frac{1}{2} \left(\alpha_1^2 (1)(1)(2) + \alpha_1 \alpha_2 (1)(-1)(4) + \alpha_2 \alpha_1 (-1)(1)(4) + \alpha_2^2 (-1)(-1)(10) \right) \right] \\ &= \max \left[(\alpha_1 + \alpha_2) - \frac{1}{2} (2\alpha_1^2 - 4\alpha_1 \alpha_2 - 4\alpha_2 \alpha_1 + 10\alpha_2^2) \right] \\ &= \max \left[(\alpha_1 + \alpha_2) - \frac{1}{2} (2\alpha_1^2 - 8\alpha_1 \alpha_2 + 10\alpha_2^2) \right] \\ &= \max [-\alpha_1^2 + 5\alpha_1 \alpha_2 - 5\alpha_2^2 + \alpha_1 + \alpha_2] \end{aligned}$$

Dual objective (simplified)

$$\max_{\alpha_1, \alpha_2} [-\alpha_1^2 + 5\alpha_1 \alpha_2 - 5\alpha_2^2 + \alpha_1 + \alpha_2]$$

Step 2 — Apply the Constraints

Constraint 1 (zero-sum):

$$\sum_{i=1}^2 \alpha_i y_i = 0 \implies \alpha_1 (1) + \alpha_2 (-1) = 0 \implies \boxed{\alpha_1 = \alpha_2}$$

Constraint 2 (box):

$$0 \leq \alpha_i \leq C = 5 \implies 0 \leq \alpha_1, \alpha_2 \leq 5$$

Step 3 — Substitute Constraint into Objective

Since $\alpha_1 = \alpha_2 \equiv \alpha$, the objective becomes:

$$f(\alpha) = -\alpha^2 + 5\alpha^2 - 5\alpha^2 + \alpha + \alpha = -\alpha^2 + 2\alpha$$

Maximise $f(\alpha) = -\alpha^2 + 2\alpha$ over $0 \leq \alpha \leq 5$:

$$f'(\alpha) = -2\alpha + 2 = 0 \implies \alpha^* = 1$$

Check boundary values:

$$f(0) = 0, \quad f(1) = -(1)^2 + 2(1) = 1, \quad f(5) = -(5)^2 + 2(5) = -15$$

Maximum is at $\alpha = 1$, giving $f(\alpha) = 1$.

Optimal dual variables

$$\alpha_1^* = 1, \quad \alpha_2^* = 1$$

Both points are support vectors.

Step 4 — Recover the Bias b

Use support vector $x_1 = 1, y_1 = +1$:

$$y_1 = \sum_{j=1}^2 \alpha_j y_j k(x_j, x_1) + b$$

$$1 = \alpha_1 y_1 k(x_1, x_1) + \alpha_2 y_2 k(x_2, x_1) + b$$

$$1 = (1)(1)(2) + (1)(-1)(4) + b = 2 - 4 + b = -2 + b$$

$$b = 3$$

Step 5 — Discriminant Function

The general discriminant function (sum over S support vectors):

$$f(z) = \sum_{j=1}^S \alpha_{t_j} y_{t_j} k(x_{t_j}, z) + b$$

With $\alpha_1 = \alpha_2 = 1$, $y_1 = 1$, $y_2 = -1$, $k(x, z) = xz + 1$, $b = 3$:

$$\begin{aligned}
 f(z) &= \alpha_1 y_1 k(x_1, z) + \alpha_2 y_2 k(x_2, z) + b \\
 &= (1)(1)(1 \cdot z + 1) + (1)(-1)(3 \cdot z + 1) + 3 \\
 &= (z + 1) - (3z + 1) + 3 \\
 &= z + 1 - 3z - 1 + 3 \\
 &= -2z + 3
 \end{aligned}$$

Step 6 — Verify Training Points

Using $f(z) = -2z + 3$ (course-note convention):

Point	z	$f(z) = -2z + 3$	Sign	Predicted class
$x_1 = 1$	1	$-2(1) + 3 = 1$	+	+1 ✓
$x_2 = 3$	3	$-2(3) + 3 = -3$	-	-1 ✓

Step 7 — Classify Test Point $z = 2$

$$f(2) = -2(2) + 3 = -1$$

Since $f(2) = -1 < 0$:

Final Answer

$$f(2) = -1 < 0 \implies \text{Output class} = -1$$

For test input $z = 2$, the SVM classifies it as the **Negative class** ($y = -1$).